

ABDUL-SOBUR AYINDE

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Portfolio: soburayinde.tech

PROFILE

Undergraduate engineer building applied AI systems for real-world problems in African public services. First author of MamaCare AI, a peer-reviewed GeoAI framework for maternal healthcare accessibility presented at the ICML 2026 Workshop on Machine Learning Research Addressing Challenges Faced by Muslim Communities, and co-author of an accepted book chapter applying geospatial intelligence to maternal health planning in Northern Nigeria. Comfortable across the full research pipeline: data collection, model development, evaluation, and the deployment and monitoring infrastructure that keeps a model trustworthy after launch. Two years of hands-on work across machine learning, backend engineering, geospatial analysis, and technical storytelling.

RESEARCH

MamaCare AI: A GeoAI Framework for Maternal Healthcare Accessibility Analysis in Underserved Communities *First author*

· ICML 2026 Workshop MusIML, Poster Track 1 · Published June 2026

OpenReview: <https://openreview.net/pdf?id=apfDp825y4>

- Built a spatial accessibility analysis across 1,510 grid points in Kebbi State, integrating GRID3 facility records (1,207 facilities), WorldPop population estimates, and Nigeria DHS health indicators.
- Applied DBSCAN spatial clustering to isolate priority underserved population zones, achieving a silhouette score of 0.83.
- Developed a composite risk scoring model across all 21 Local Government Areas, identifying Argungu, Augie, and Dandi as critical intervention priorities.
- Proposed a phased AI-assisted mobile clinic deployment strategy reaching 310,000 women across three priority LGAs. The paper estimates that roughly 62% of women of reproductive age in Kebbi State live more than 5 km from the nearest health facility.
- Solo-authored, non-archival workshop submission, eligible for journal or full-conference extension.

Geospatial Intelligence for Targeted Maternal Health Interventions in Northern Nigeria: A Case Study of Kebbi State *Co-author with Mr. Onyemaachi John Nnamani*

· Accepted book chapter · 2026

- Explored how geospatial intelligence and AI-driven spatial analysis can support targeted maternal health interventions, improve resource allocation, and strengthen healthcare access in underserved regions.

Forthcoming in Sustainable and Multidisciplinary Advanced Research Trends for AI-Driven Future in Africa

Using AI and Machine Learning for Land Use Classification in Zimbabwe *Co-author*

· Published via ResearchGate · 2025

ResearchGate: https://www.researchgate.net/publication/395703408_Using_AI_and_Machine_learning_for_land_use_classification_in_Zimbabwe_Enhancing_Geospatial_Governance_for_SDG_Achievement

- Applied machine learning to satellite land-use classification in support of geospatial governance and SDG measurement. Manuscript extended from this work is in preparation.

TECHNICAL PORTFOLIO

Code sample for this application: <https://croesusml.vercel.app/proof.html>

Documented ML systems portfolio covering evaluation, safety, and reliability infrastructure, with working evidence rather than descriptions.

- FraudShield: evaluation framework for a fraud detection model, with reproducible metrics and error analysis.
- SecureRAG: retrieval-augmented generation system hardened against prompt injection and retrieval poisoning, validated against a 120+ case adversarial security test suite.
- Drift monitoring: production runbook using population stability index thresholds and Kolmogorov–Smirnov tests for detecting distribution shift.
- RAG evaluation dashboard: retrieval quality, groundedness, and answer accuracy tracked over time.
- Cost and latency reporting: per-query performance and spend instrumentation.
- Governance artifacts: model card, dataset datasheet, and written incident postmortem.

PROJECTS

GeoBot *Python · FastAPI · LLM APIs*

- Conversational assistant for geospatial queries, built to make location data accessible to non-specialists.

IGeocode *Python · REST APIs · PostgreSQL*

- Geocoding service converting informal Nigerian address descriptions into usable coordinates.

FUTA Student Accommodation Platform *Python · Flask · MongoDB*

- Web platform connecting FUTA students with verified realtors and potential roommates. Handles property listings, student profiles, and roommate preference matching, with the goal of shortening housing search time and reducing exposure to unverified listings.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL

Machine Learning: TensorFlow, Scikit-learn, Pandas, NumPy, OpenCV, spatial clustering (DBSCAN), model evaluation, drift monitoring

Backend: Flask, FastAPI, REST API design

Databases: PostgreSQL, MySQL, MongoDB

Geospatial: Raster and vector analysis, land-use classification, spatial accessibility modelling

Tooling: Git, GitHub, Docker (working knowledge), Matplotlib, Seaborn

Other: VFX and video editing, Agile/Scrum

LEADERSHIP & EXTRACURRICULAR

Host, Before They Graduate *Student storytelling series*

- Interview-based series documenting student experiences and ambitions. Produced end to end through research, filming, editing, and publication.

Tech Off Hackathon *Participant*

- Delivered a working solution within a 2-hour compressed build challenge.

Departmental and International Student Programs *Program support and coordination*

- Held leadership roles supporting departmental initiatives and programs for international students, working across teams and cultural contexts.

Department of Surveying and Geoinformatics, FUTA *Student leadership*

- Previously served as Vice President and Assistant General Secretary, supporting student programs, communication, documentation, and collaborative initiatives.

EDUCATION

Federal University of Technology, Akure (FUTA)

B.Tech, Surveying and Geoinformatics — Ongoing (2023–Present)

Robotics and Artificial Intelligence Nigeria (RAIN)

Artificial Intelligence and Machine Learning (Python), 2024

Orange VFX Academy

Visual Effects (VFX) 2025

REFERENCES

Available on request.